Yawar Hussain

Geophysicist - Rock Mass & Ground Deformation Monitoring

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Google Scholar: Citations - 1779, H-Index - 21)

CORE EXPERTISE

- Ground-deformation and infrastructure monitoring via distributed fibre-optic sensing (DAS, DTS), borehole strainmeter, and inertial sensors (Geophones, Seismometers, Nodes): hanging-wall, slope, and dam-crest systems.
- Seismogenic and earthquake-risk assessment: engineering-geological and seismic risk studies at Aegina Kolonna, Greece, and the Le Capore and Pescheria springs, Rome.
- Subsidence and ground-vibration assessment: site-response and infrastructure-exposure studies in Mexico City and Arequipa, Peru.
- Seismic rock-mass characterisation: ambient-noise interferometry, dv/v and resonance-frequency monitoring, MASW dispersion analysis.
- Subsurface and geohazard characterisation: GPR, ERT, AMT/MT.
- Near-surface geophysical groundwater exploration and characterisation.
- Full survey lifecycle: design, field acquisition, QA/QC, processing, modelling, and interpretation.

EMPLOYMENT

Specialist Infrastructure Hazard Monitoring (01/2026 to Present), HydroResearch, Stockholm, Sweden

- Supporting the expansion of fibre-optic monitoring technologies (DTS, DSS, DAS) through integrated field measurements, system operation, and data evaluation.
- Running acoustic monitoring, quality control, alarm follow-up, and technical reporting - end-to-end dam monitoring from measurement to interpretation.

Research Associate (11/2024 to 06/2025), Department of Sustainable Earth Systems Sciences, University of Texas at Dallas, USA

- Applied near-surface geophysics and modelling to the Pecos Valley Aquifer System, Texas, configuring PulseEKKO GPR systems for field operations and instructional demonstrations.
- Taught a graduate-level course on structural health monitoring and earthquake-science communication; completed an advanced Postdoctoral Teaching Certificate.

Post-doctoral Researcher (01/2023 to 11/2024), Department of Earth Science, 'Sapienza' University of Rome, Italy

- Analysed data from multiparametric monitoring systems for slope-instability detection affecting strategic infrastructure and cultural heritage.
- Delivered structural stability risk assessment, monitoring of the Le Capore and Pescheria hydrostructures, a multi-hazard assessment framework for a historical landslide, and seismic monitoring of the Acuto Field Lab.

Visiting Professor (04/2024), University of Brasília, Brazil

- Applied shallow geophysical techniques, principally GPR, to geomorphological targets in tropical environments.

- Taught a three-credit-hour course on geophysical techniques for geomorphological studies, Department of Geography and Institute of Geosciences.

Post-doctoral Researcher (01/2021 to 12/2022), Department of Geology, University of Liège, Belgium

- Geophysical site-selection studies for the Einstein Telescope installation and the El Hierro mega-landslide.
- Installed signal-station, mobile, and semi-permanent broadband seismometer arrays (FDSN network 5B).

Post-doctoral Researcher (01/2020 to 12/2020), Environmental Engineering and Earth Sciences, Clemson University, USA

- Recorded and analysed deformation/strain using a state-of-the-art strainmeter during an injection test in Oklahoma.

Post-doctoral Researcher (08/2018 to 01/2020), CECAV - ICMBio, Brasília, Brazil

- Geophysical (GPR, ERT) and geotechnical characterisation of subsurface caves for geohazard assessment and karst-aquifer vulnerability.

SELECTED PROJECTS

- **Hanging Wall Monitoring with DAS** - Design and processing of a Distributed Acoustic Sensing (DAS) survey for hanging wall stability monitoring, combining passive ambient-noise interferometry (dv/v velocity monitoring, resonance-frequency tracking); authored the feasibility assessment and technical report.
- **Dam Monitoring with DAS -** Design and processing of a Distributed Acoustic Sensing (DAS) survey for embankment dam analysis with passive geophysical tools.
- Multiscale Geophysical and Petrophysical Monitoring of Coastal Cliffs and Urban Slopes in Northeastern Brazil (on going) - CNPq (Brazilian Federal Funding Agency).
- ACEA postdoctoral grant: Analysis of data from multiparametric monitoring systems for slope instability control.
- Wallonia-Brussels International (WBI) Excellence Grant: Geophysical applications for the site selection of the Einstein Telescope installation and mega landslide in El Hierro, Spain.
- CAPES postdoctoral grant: The application of geophysical and geotechnical monitoring technologies contributes to the understanding of the dynamics of mass movements.
- PhD grant: Analysis of the dynamics of an unstable artificial slope using Nanoseismic Monitoring and numerical simulation, Funding agency: FAPDF

EDUCATION

08/2014 to 09/2018

Ph.D. in Geotechnical Engineering

Department of Civil and Environmental Engineering, University of Brasilia, Brazil.

03/2012 to 03/2014

M.Phil in HydroGeophysics

Department of Earth Sciences, Quaid-i-Azam University, Islamabad, Pakistan.

02/2009 to 02/2011

Masters in Geophysics

Department of Earth Sciences, Quaid-i-Azam University, Islamabad, Pakistan.

06/2007 to 08/2009

Bachelor's Degree

University of the Punjab, Lahore, Pakistan.

EDITORIAL CONTRIBUTIONS

- **Handling Editor:** Seismica.
- **Associate Editor:** International Journal of Economic and Environmental Geology (IJEEG) (2026 - present).
- **Topic Editor:** (currently accepting papers) Role of Innovative Near Surface Geophysical Techniques in Understanding Geological Instabilities: A Geohazard Perspective, Discover Geosciences, Springer.
- **Topic Editor:** (completed) Editorial: Near-surface geophysics in Latin American contexts: its applications, education, and societal perspectives as a whole, Frontiers in Earth Science.
- **Young Editorial board member:** Natural Hazards Research, URL; Discover Hazards.
- **Reviewer:** Geoscience Communication.

SELECTED PUBLICATIONS

Geophysics in Geohazards:

- Marano, et al. Hussain, et al. (internal revision) Integrated geophysical and remote sensing techniques to constrain seismic response and activity of the urban landslide at San Vito Romano (Italy)
- Xavier, et al. Hussain, et al. (2025) Impact Assessment of Infrastructures Exposure to Weak Motion Vibrations in Arequipa, Peru, Brazilian Journal of Geophysics, 43(2):27
- Sokolicek, et al. Hussain, et al. (2025) Engineering-geological and geophysical surveys for archaeological risk assessment and previous mitigation measures at the CH-site of Aegina Kolonna, Greece, Italian Journal of Engineering Geology and Environment
- Innocenti, et al. Hussain et al (2025) Geophysical reconnaissance of the Pays de Herve landslides: case study of three active paleo-slides in a dissected tableland, Landslides
- Antonielli, et al. Hussain, et al. (2025). Local seismic response in the historical centre of Nafplio (Greece) as a tool for seismic risk management, Natural Hazards, 121(13):15581–15611
- Cárdenas-Soto, et al. Hussain et al. (2021) Seismic ambient noise tomography to retrieve near-surface properties in soils with significant 3D lateral heterogeneity. The case of Quinta Colorada building in Chapultepec, Mexico, Natural Hazards
- Qian, et al. Hussain., …, et al. (2024). Impact of environmental stressors on jointed rock cliffs by acoustic emission sensing: preliminary findings from the Acuto Field Laboratory (Central Italy), Rock Mechanics and Rock Engineering
- Hussain, et al. (2022). Review on the Geophysical and UAV-Based Methods Applied to Landslides. Remote Sensing
- Klimeš, Hussain, et al. (2023). New Insights into the Internal Structures and Geotechnical Rock Properties of the Giant San Andrés Landslide, El Hierro Island, Spain. Remote Sensing
- Hamza, et al. Hussain, et al. (2019). Geohazard Assessment of Landslides in South Brazil: Case Study. Geotechnical and Geological Engineering, URL
- Hussain et al. (2019). Multiple Geophysical Techniques for Investigation and Monitoring of Sobradinho Landslide, Brazil. Sustainability
- Hussain et al. (2019). Monitoring of Sobradinho landslide (Brasília, Brazil) and a prototype vertical slope by time-lapse interferometry. Brazilian Journal of Geology

- Hussain et al. (2020). Seasonal variation in Rayleigh wave ellipticity as a possible indicator of earthflow mobility: a case study of Sobradinho landslide compared with pile load testing, Earth Sciences Research Journal
- Hussain et al. (2019). Typological analysis of slidequakes emitted from landslides: experiments on an expander body pile and Sobradinho landslide (Brasilia, Brazil). REM–International Engineering Journal
- Hamza, et al. Hussain et al. (2022). Evaluation of the subgrade reaction for closely-spaced shallow foundations on sand, Geotechnical Engineering (Proceedings of the ICE
- Hussain, et al. (2020). Characterization of Sobradinho landslide in fluvial valley using MASW and ERT methods. REM–International Engineering Journal
- Hussain et al. (2020). Seismic signature of mudflow tremor resulted from Brumadinho (Brazil) tailings dam failure. REM–International Engineering Journal
- Hussain et al. (2019). Introductory review of potential applications of nanoseismic monitoring in seismic energy characterization. Journal of Engineering Research
- Hussain, et al. (2023). Seismic Signatures and Site Characterization of an Intermittent Stream in Dry and Flood Conditions: An Implication for Soil Losses and Landslide Triggering, Environmental Earth Science

SELECTED CONFERENCES

- Human-Induced Seismicity Front in Texas and Risk Preparedness: University Students Exercise, 2025 NSF GAGE/SAGE Community Science Workshop, Bloomington, USA (poster)
- Geohazard Identification at Le Capore Spring, Central Apennines, Italy, AIGAA Napoli 2024 (talk)
- Passive seismic monitoring of a jointed rock mass at Acuto Field Lab (Italy), Eurock 2025, Norway (presentation by Matteo)
- Multi-scale monitoring of rock mass deformations in a mid-latitude climatic context through a Fiber Bragg Grating array, Eurock 2025, Norway
- Integrated geotechnical, acoustic and microseismic monitoring of a jointed rock-wall prone to block failures: results from the Acuto Field Lab (Italy), EUROCK-2024, Spain
- Landslide detection and monitoring by integrating electrical, seismic and interferometric techniques in a multi-hazard perspective: the case of San Vito Romano (RM), GNGTS 2025, Italy
- Slope-scale deformation process interacting with hydrostructure: insights from geotechnical monitoring, geophysical investigation and remotely sensed observations, XIVth International Symposium on Landslides-2024, France